

Diwali Sales Analysis project

Load The dataset and the libraries

```
In [54]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns

df=pd.read_csv(r'C:\Users\abrar\Downloads\Python_Diwali_Sales_Analysis\Python_Diwali_Sales_Analysis\Diwali Sales Data\Diwali Sales Data.csv')
df
```

Out[54]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	
...	...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	

11251 rows × 15 columns



Basic data cleaning

Check the first 5 rows of the dataset

In [55]: `df.head()`

Out[55]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Categ
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	A
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	A
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	A
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	A
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	A

Check the basic info of the dataset

In [56]: `df.info()`

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   User_ID               11251 non-null  int64  
1   Cust_name             11251 non-null  object  
2   Product_ID           11251 non-null  object  
3   Gender                11251 non-null  object  
4   Age Group             11251 non-null  object  
5   Age                   11251 non-null  int64  
6   Marital_Status        11251 non-null  int64  
7   State                 11251 non-null  object  
8   Zone                  11251 non-null  object  
9   Occupation            11251 non-null  object  
10  Product_Category      11251 non-null  object  
11  Orders                11251 non-null  int64  
12  Amount                11239 non-null  float64 
13  Status                0 non-null      float64 
14  unnamed1              0 non-null      float64 
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB

```

Drop the unrelated or blanks columns from the dataset

```
In [57]: df.drop(['Status', 'unnamed1'],axis=1,inplace=True,errors="ignore")
df
```

Out[57]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	
...	...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	

11251 rows × 13 columns



Check the null values

```
In [58]: pd.isnull(df)
```

Out[58]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Category	Orc
0	False	False	False	False	False	False	False	False	False	False	False	F
1	False	False	False	False	False	False	False	False	False	False	False	F
2	False	False	False	False	False	False	False	False	False	False	False	F
3	False	False	False	False	False	False	False	False	False	False	False	F
4	False	False	False	False	False	False	False	False	False	False	False	F
...	...	...	...	...	...	...	...	...	...	...	...	...
11246	False	False	False	False	False	False	False	False	False	False	False	F
11247	False	False	False	False	False	False	False	False	False	False	False	F
11248	False	False	False	False	False	False	False	False	False	False	False	F
11249	False	False	False	False	False	False	False	False	False	False	False	F
11250	False	False	False	False	False	False	False	False	False	False	False	F

11251 rows × 13 columns

In [59]: `pd.isnull(df).sum()`

```
Out[59]: User_ID      0
         Cust_name    0
         Product_ID   0
         Gender        0
         Age Group     0
         Age           0
         Marital_Status 0
         State         0
         Zone          0
         Occupation    0
         Product_Category 0
         Orders        0
         Amount        12
         dtype: int64
```

```
In [60]: df.shape
```

```
Out[60]: (11251, 13)
```

Drop the null values

```
In [61]: df.dropna(inplace=True)
```

```
In [62]: df.shape
```

```
Out[62]: (11239, 13)
```

Change the data type

```
In [63]: df["Amount"] = df['Amount'].astype(int)
         df['Amount'].dtypes
```

```
Out[63]: dtype('int64')
```

Check Column in the dataset

```
In [64]: df.columns
```

```
Out[64]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
              'Orders', 'Amount'],
              dtype='object')
```

Rename a column

```
In [65]: df.rename(columns={'Marital_Status':'Marital_Status'},inplace=False)
```

Out[65]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	
...	...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical	
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare	
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile	
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture	
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare	

11239 rows × 13 columns



Description of the data in the dataframe

```
In [66]: df.describe()
```

Out[66]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [67]: `df[['Age', 'Orders', 'Amount']].describe()`

Out[67]:

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Exploratory Data Analysis

Gender analysis

In [68]: `df.head()`

Out[68]:

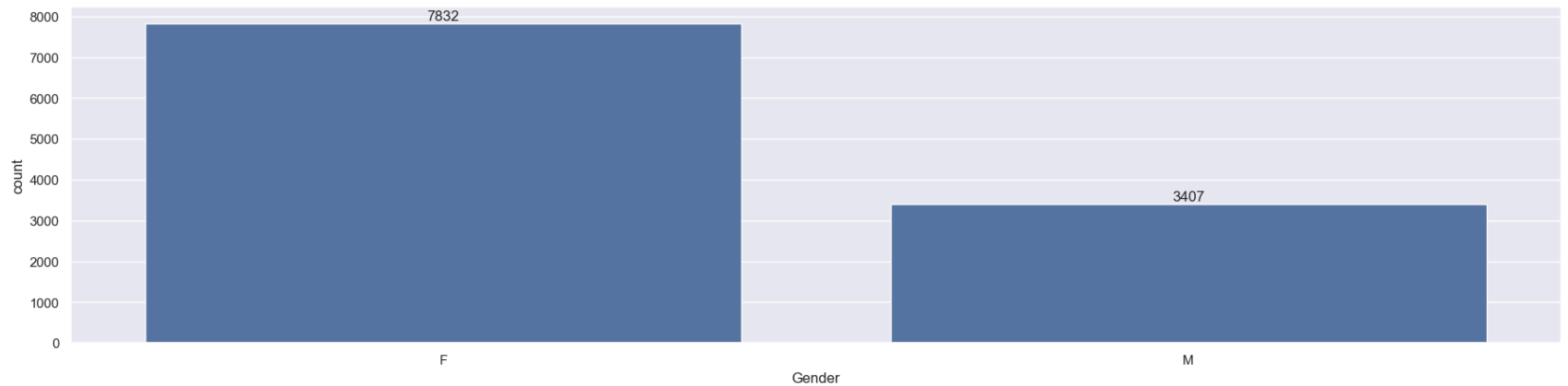
	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Marital_Status	State	Zone	Occupation	Product_Categ
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare	A
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt	A
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile	A
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction	A
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing	A

In [69]: `df.columns`

Out[69]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
'Orders', 'Amount'],
dtype='object')

In [70]: `ax=sns.countplot(x='Gender',data=df)`

```
for bars in ax.containers:
    ax.bar_label(bars)
```



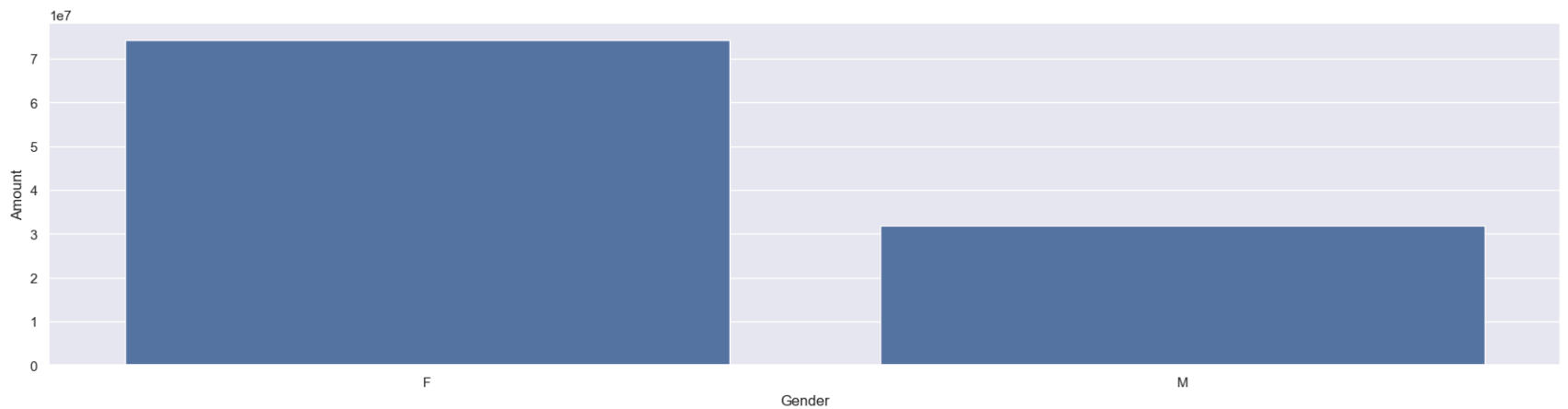
```
In [71]: sales_gen=df.groupby(['Gender'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=False)  
sales_gen
```

```
Out[71]:
```

	Gender	Amount
0	F	74335853
1	M	31913276

```
In [72]: sns.barplot(x='Gender',y='Amount',data=sales_gen)
```

```
Out[72]: <Axes: xlabel='Gender', ylabel='Amount'>
```



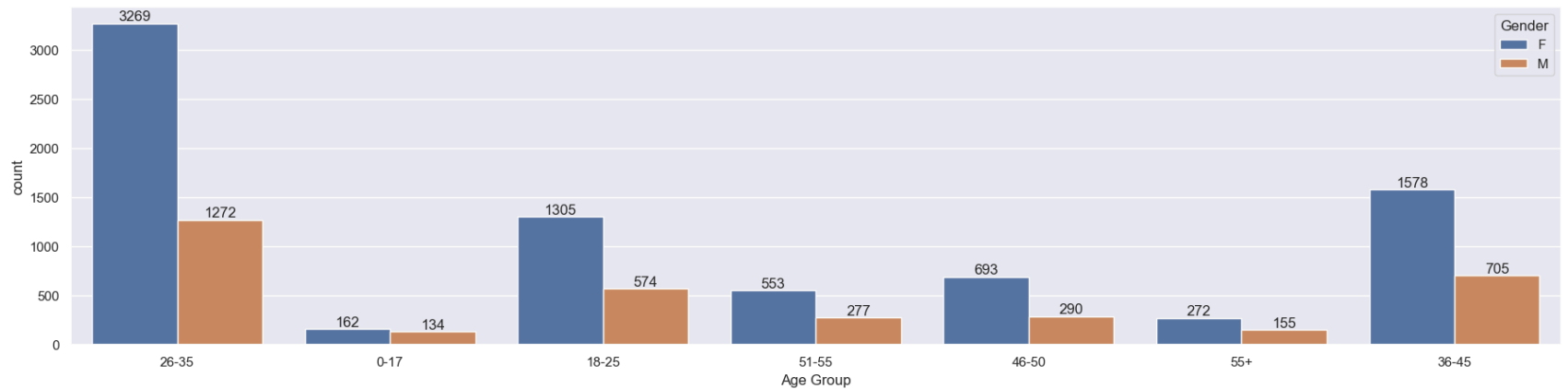
From above the graph we can see that most of the buyers are female and the purchasing power of the female is greater than the men

Age Analysis

```
In [73]: df.columns
```

```
Out[73]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
              'Orders', 'Amount'],  
            dtype='object')
```

```
In [74]: ax=sns.countplot(data=df,x='Age Group',hue='Gender')  
  
for bars in ax.containers:  
    ax.bar_label(bars)
```



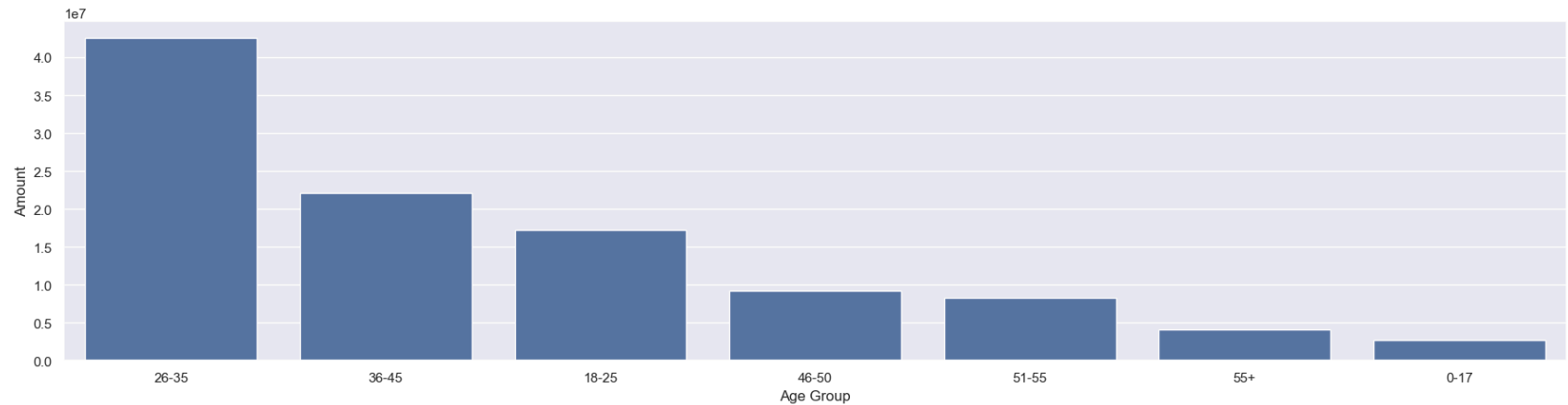
```
In [75]: sales_age=df.groupby(['Age Group'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=False)  
sales_age
```

```
Out[75]:
```

	Age Group	Amount
2	26-35	42613442
3	36-45	22144994
1	18-25	17240732
4	46-50	9207844
5	51-55	8261477
6	55+	4080987
0	0-17	2699653

```
In [76]: sns.barplot(x='Age Group',y='Amount',data=sales_age)
```

```
Out[76]: <Axes: xlabel='Age Group', ylabel='Amount'>
```



From above the graph we can see that most of the buyers are 26-35 years old female

State Analysis

```
In [77]: df.columns
```

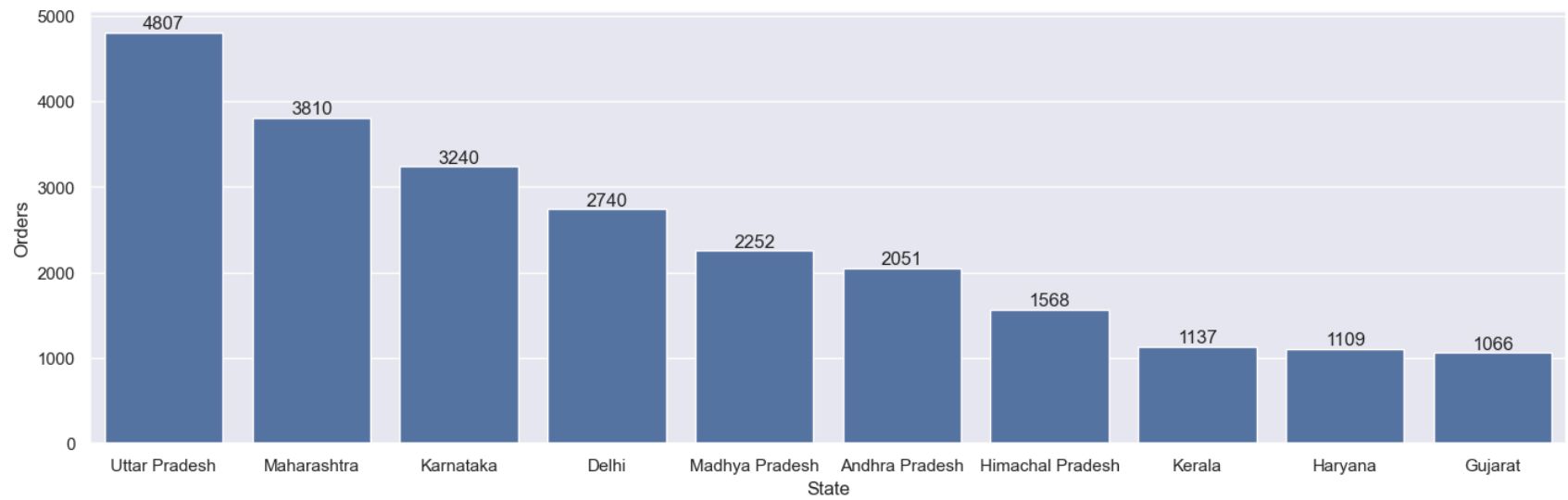
```
Out[77]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
              'Orders', 'Amount'],
              dtype='object')
```

```
In [78]: #sales vs orders
sales_state=df.groupby(['State'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False).head(10)
sales_state
```

```
Out[78]:
```

	State	Orders
14	Uttar Pradesh	4807
10	Maharashtra	3810
7	Karnataka	3240
2	Delhi	2740
9	Madhya Pradesh	2252
0	Andhra Pradesh	2051
5	Himachal Pradesh	1568
8	Kerala	1137
4	Haryana	1109
3	Gujarat	1066

```
In [79]: sns.set(rc={'figure.figsize':(17,5)})
ax=sns.barplot(x='State',y='Orders',data=sales_state)
for bars in ax.containers:
    ax.bar_label(bars)
```



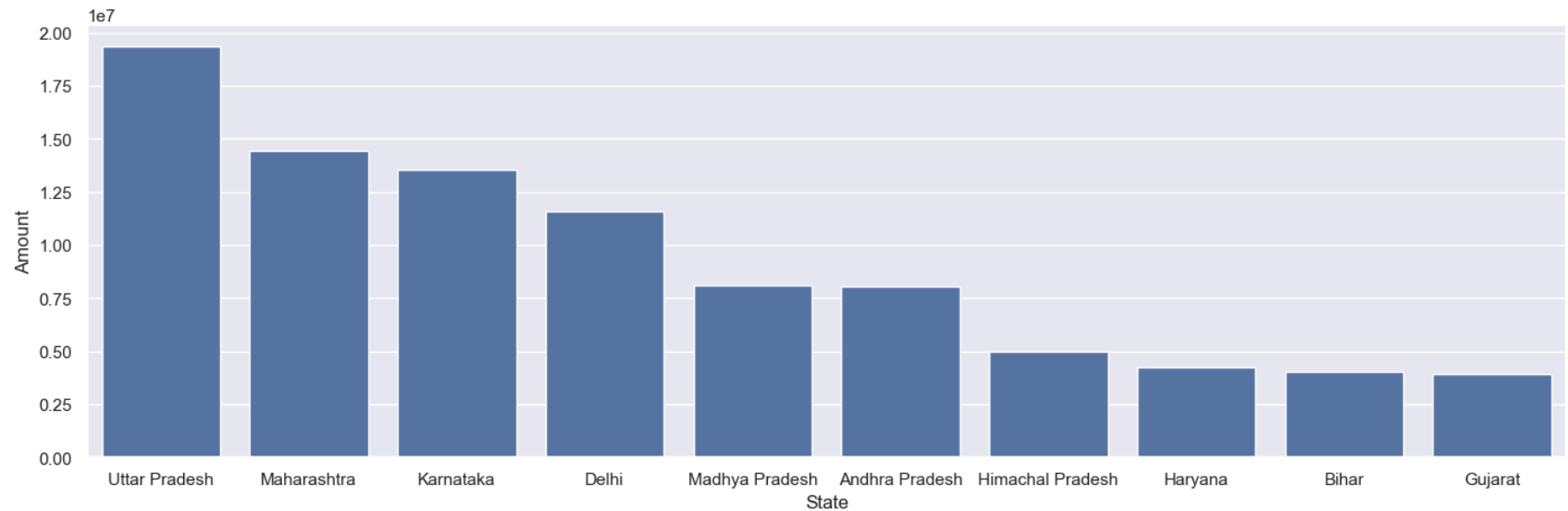
```
In [80]: # sales vs amount
sales_state= df.groupby(['State'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=False).head(10)
sales_state
```

Out[80]:

	State	Amount
14	Uttar Pradesh	19374968
10	Maharashtra	14427543
7	Karnataka	13523540
2	Delhi	11603818
9	Madhya Pradesh	8101142
0	Andhra Pradesh	8037146
5	Himachal Pradesh	4963368
4	Haryana	4220175
1	Bihar	4022757
3	Gujarat	3946082

```
In [81]: sns.set(rc={'figure.figsize':(17,5)})  
sns.barplot(x='State',y='Amount',data=sales_state)
```

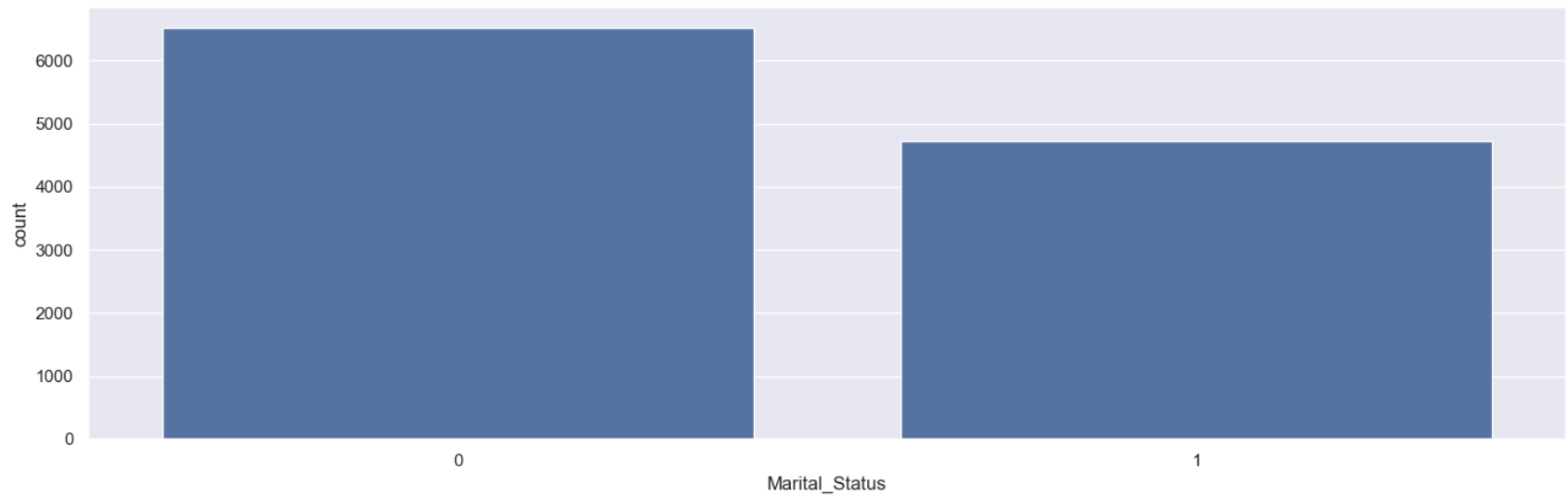
```
Out[81]: <Axes: xlabel='State', ylabel='Amount'>
```



From above the graph we can see that most of the orders and total sales/amount are from the Uttar Pradesh, Maharashtra and Karnataka.

Marital Status Analysis

```
In [82]: sns.countplot(data=df,x='Marital_Status')  
sns.set(rc={'figure.figsize':(7,5)})
```



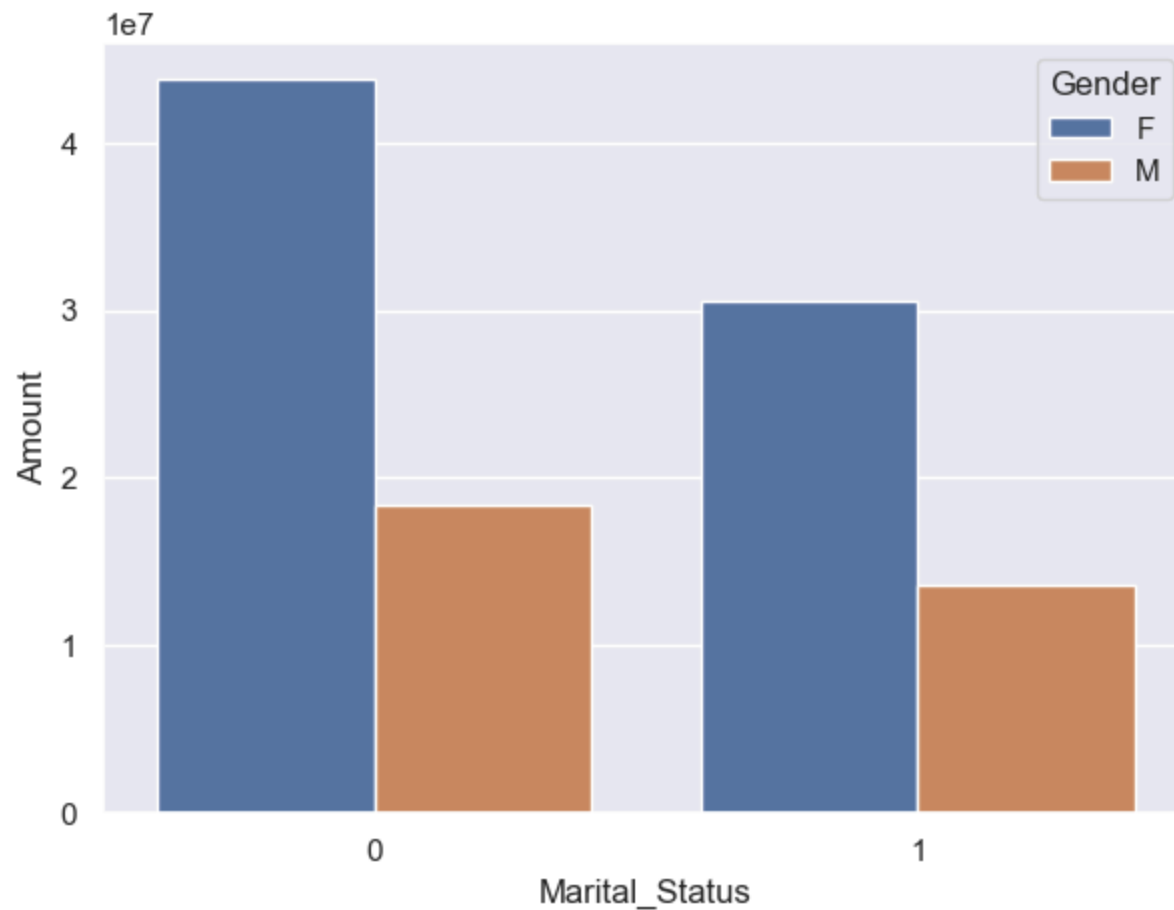
```
In [83]: sales_marit=df.groupby(['Marital_Status','Gender'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=sales_marit)
```

```
Out[83]:
```

	Marital_Status	Gender	Amount
0	0	F	43786646
2	1	F	30549207
1	0	M	18338738
3	1	M	13574538

```
In [84]: sns.barplot(x='Marital_Status',y='Amount',data=sales_marit,hue='Gender')
plt.figure(dpi=500)
```

```
Out[84]: <Figure size 3500x2500 with 0 Axes>
```



<Figure size 3500x2500 with 0 Axes>

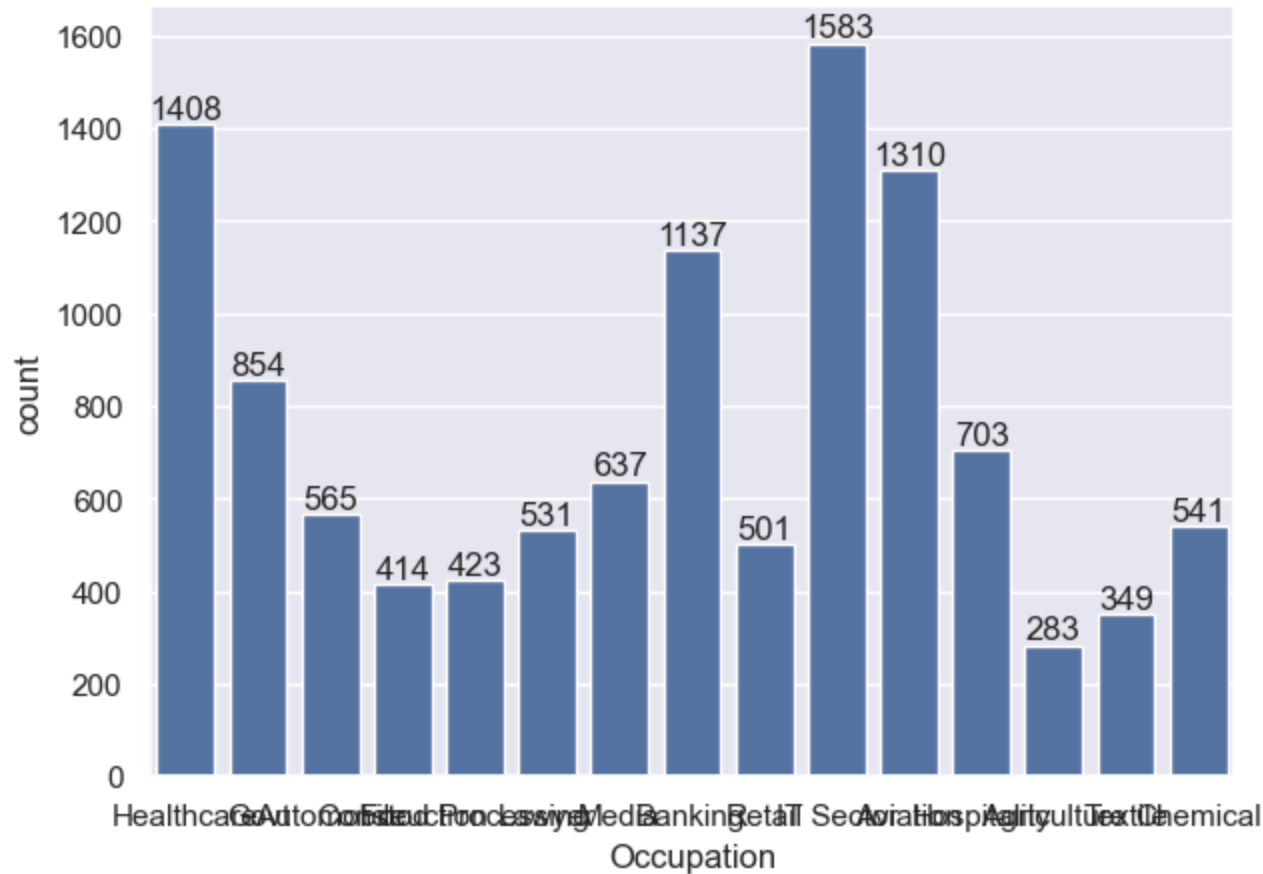
From above graph we can see that most of the buyers are married women and they have also high purchasing power.

Occupation Analysis

```
In [85]: df.columns
```

```
Out[85]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
              'Orders', 'Amount'],  
             dtype='object')
```

```
In [86]: ax=sns.countplot(x='Occupation',data=df)
sns.set(rc={'figure.figsize':(22,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



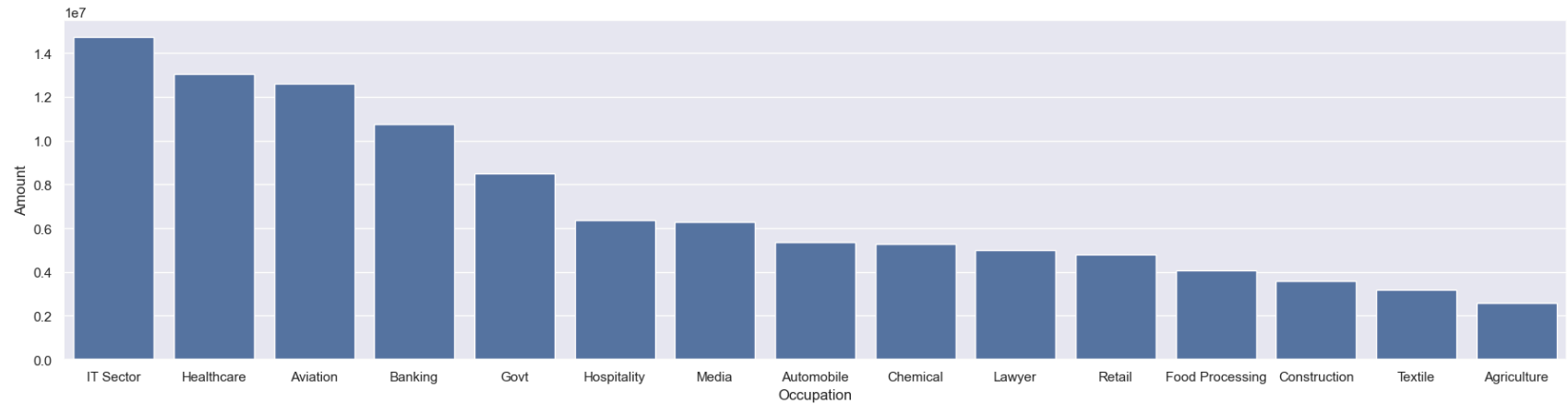
```
In [87]: sales_occupation=df.groupby(['Occupation'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=False)
sales_occupation
```

Out[87]:

	Occupation	Amount
10	IT Sector	14755079
8	Healthcare	13034586
2	Aviation	12602298
3	Banking	10770610
7	Govt	8517212
9	Hospitality	6376405
12	Media	6295832
1	Automobile	5368596
4	Chemical	5297436
11	Lawyer	4981665
13	Retail	4783170
6	Food Processing	4070670
5	Construction	3597511
14	Textile	3204972
0	Agriculture	2593087

```
In [88]: sns.barplot(x='Occupation',y='Amount',data=sales_occupation)
```

```
Out[88]: <Axes: xlabel='Occupation', ylabel='Amount'>
```



From above the graph we can see that most of the buyers working in the IT sector, healthcare and aviation.

Product category analysis

```
In [89]: df.columns
```

```
Out[89]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
               'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
               'Orders', 'Amount'],
              dtype='object')
```

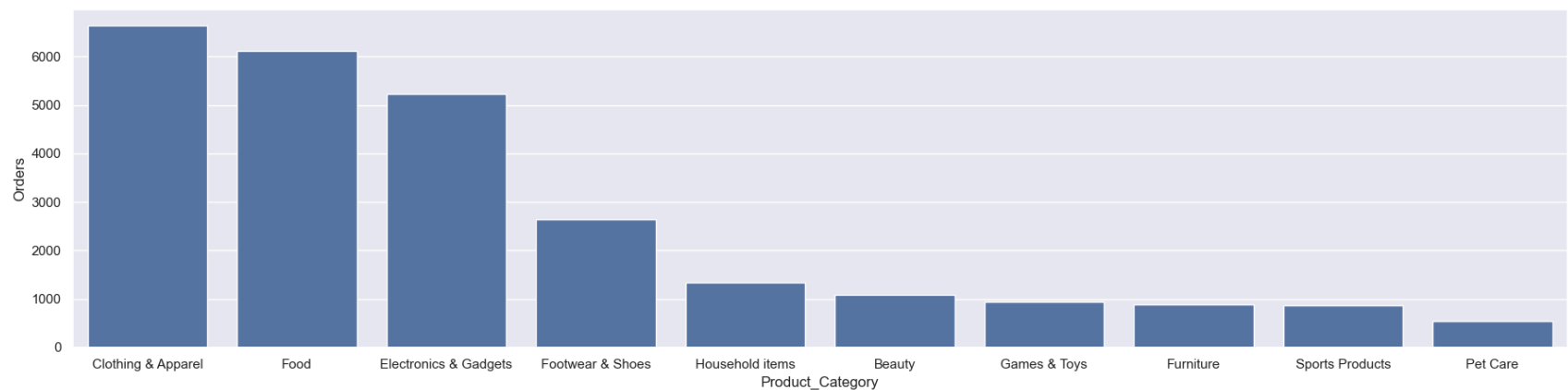
```
In [92]: product_category=df.groupby(['Product_Category'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False)
product_category
```

Out[92]:

	Product_Category	Orders
3	Clothing & Apparel	6634
6	Food	6110
5	Electronics & Gadgets	5226
7	Footwear & Shoes	2646
11	Household items	1331
1	Beauty	1086
9	Games & Toys	940
8	Furniture	889
14	Sports Products	870
13	Pet Care	536

```
In [94]: sns.barplot(x='Product_Category',y='Orders',data=product_category)
```

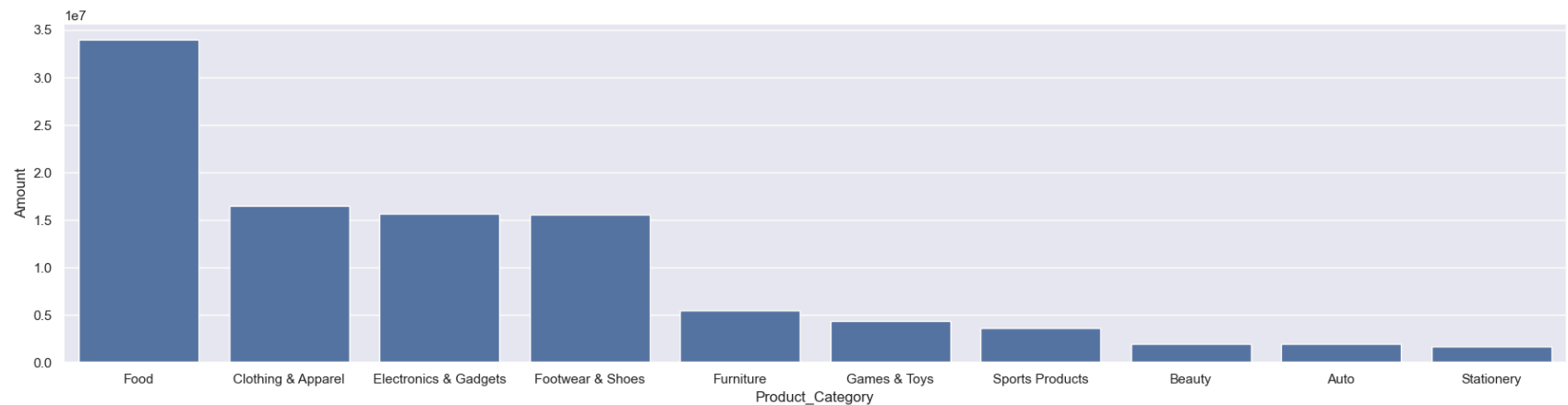
```
Out[94]: <Axes: xlabel='Product_Category', ylabel='Orders'>
```



```
In [95]: product_category=df.groupby(['Product_Category'],as_index=False)['Amount'].sum().sort_values(by='Amount',ascending=False)  
product_category
```

Out[95]:

	Product_Category	Amount
6	Food	33933883
3	Clothing & Apparel	16495019
5	Electronics & Gadgets	15643846
7	Footwear & Shoes	15575209
8	Furniture	5440051
9	Games & Toys	4331694
14	Sports Products	3635933
1	Beauty	1959484
0	Auto	1958609
15	Stationery	1676051

In [96]: `sns.barplot(x='Product_Category',y='Amount',data=product_category)`Out[96]: `<Axes: xlabel='Product_Category', ylabel='Amount'>`

From avobe the graph we can see that most sold product are food, clothing and electronics.

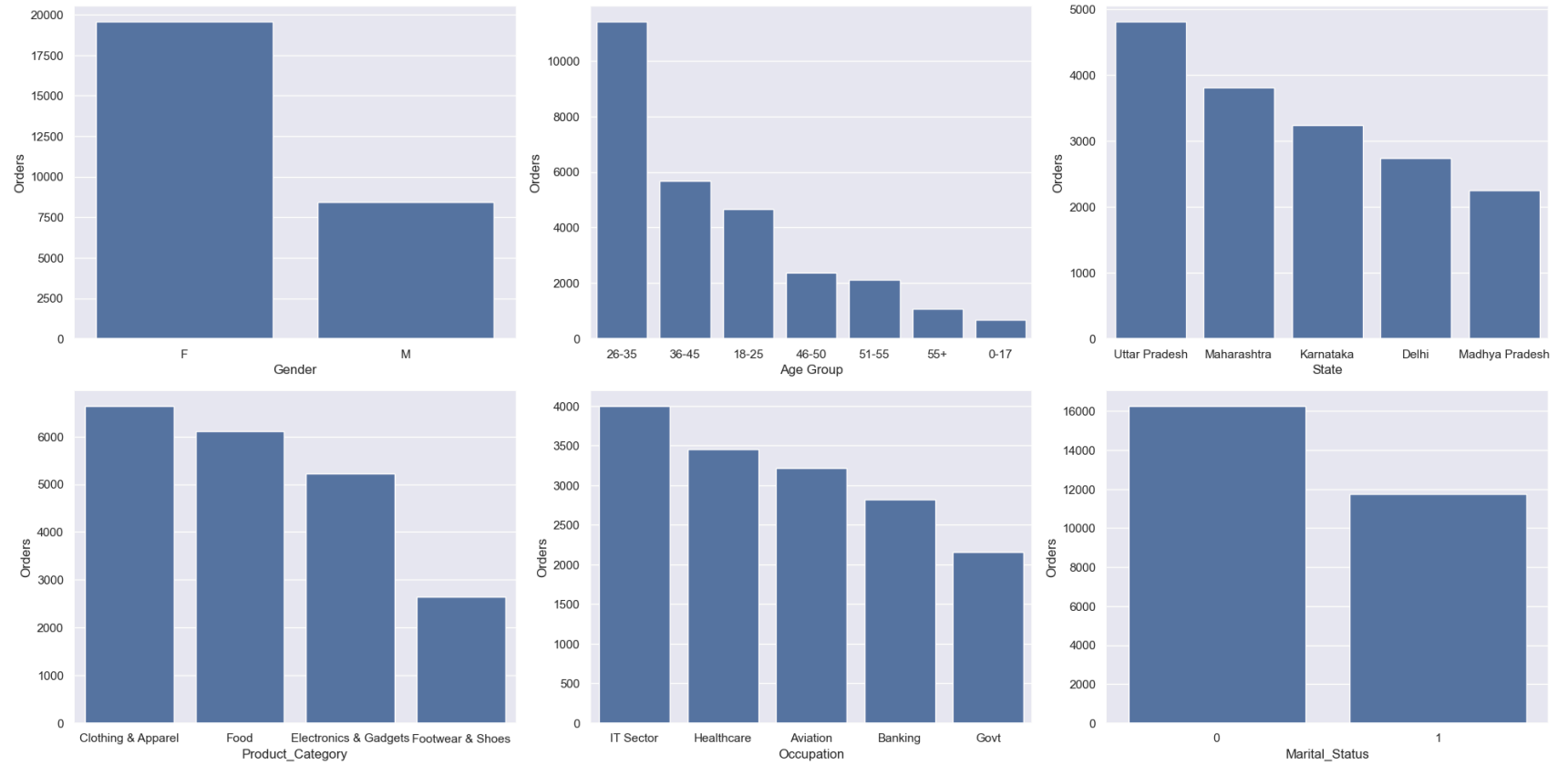
Customer Order Analysis Across Different Demographic Segments

```
In [98]: df.columns
```

```
Out[98]: Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',  
              'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',  
              'Orders', 'Amount'],  
              dtype='object')
```

```
In [105... gender=df.groupby(['Gender'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False)  
age=df.groupby(['Age Group'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False)  
marit=df.groupby(['Marital_Status'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False)  
state=df.groupby(['State'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False).head(5)  
occupation=df.groupby(['Occupation'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False).head(5)  
category=df.groupby(['Product_Category'],as_index=False)['Orders'].sum().sort_values(by='Orders',ascending=False).head(5)
```

```
In [108... # Create subplot layout  
fig, axes = plt.subplots(2, 3, figsize=(20,10))  
  
sns.barplot(x='Gender', y='Orders', data=gender, ax=axes[0,0])  
sns.barplot(x='Age Group', y='Orders', data=age, ax=axes[0,1])  
sns.barplot(x='State', y='Orders', data=state, ax=axes[0,2])  
  
sns.barplot(x='Product_Category', y='Orders', data=category, ax=axes[1,0])  
sns.barplot(x='Occupation', y='Orders', data=occupation, ax=axes[1,1])  
sns.barplot(x='Marital_Status', y='Orders', data=marit, ax=axes[1,2])  
  
plt.tight_layout()  
plt.show()
```



The analysis shows that:

1. **Female customers** placed the highest number of orders
2. **The 26–35 age group** is the most active customer segment
3. **Uttar Pradesh, Maharashtra, and Karnataka** are the strongest markets
4. **Clothing & Apparel** is the most ordered product category
5. **IT sector professionals** contributed the highest number of orders
6. **Married customers** showed relatively higher purchasing behavior

Final Business Conclusion:

During the Diwali sales period, The most valuable target customer segment appears to be female working professionals aged 26–35 from high-performing states, with strong interest in Clothing & Apparel products. Focusing marketing, inventory planning, and customer engagement on this segment could significantly improve business growth.